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Internship report

The Riemann-Hilbert correspondence

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II Prerequisites

II.1 Category theory

For the first three subsubsection, we follow the presentation by Assem in [Ass22] and try to summurize it for the purpuse of this paper. We add the precise reference for the aforementioned book to each paragraph and each result that would need a proof.
Of course, we expose the basic notions of categories, so many books contains such material.

II.1.1 First notions

a. Starting point

[[Ass22], I].

Definition II.1.

A *category* $\mathscr{C}$ is given by the data of :

1. a class $\mathrm{Ob}(\mathscr{C})$, called the *class of objects* of $\mathscr{C}$,
2. for every couple of objects (x, y) of $\mathscr{C}$, a class $\mathrm{Hom}_{\mathscr{C}}(x, y)$, called the *class of morphisms* from x to y,
3. for every triplet of objects (x, y, z) of $\mathscr{C}$, a function

$$\begin{aligned} \mathrm{comp}_{(x,y,z)} : \quad \mathrm{Hom}_{\mathscr{C}}(x, y) \times \mathrm{Hom}_{\mathscr{C}}(y, z) &\longrightarrow \mathrm{Hom}_{\mathscr{C}}(x, z) \\ (f, g) &\longmapsto g \circ f \end{aligned}$$

   satisfying the following proprerties :

   a. (*Associative law*) for every quartet of objects (w, x, y, z) of $\mathscr{C}$, if $(f, g, h) \in \mathrm{Hom}_{\mathscr{C}}(w, x) \times \mathrm{Hom}_{\mathscr{C}}(x, y) \times \mathrm{Hom}_{\mathscr{C}}(y, z)$, then $(h \circ g) \circ f = h \circ (g \circ f)$,
   b. (*Identities*) for every object x of $\mathscr{C}$, there exists an element 1_x of $\mathrm{Hom}_{\mathscr{C}}(x, x)$, so that for every couple of objects (w, y) and every morphisms $(f, g) \in \mathrm{Hom}_{\mathscr{C}}(w, x) \times \mathrm{Hom}_{\mathscr{C}}(x, y)$, then $1_x \circ f = f$ and $g \circ 1_x = g$.

   $g \circ f$ is called the *composition* of g with f.

Remark II.1.

- We will use the following notations : $x \in \mathrm{Ob}(\mathscr{C})$ if x in an object of $\mathscr{C}$, $f : x \to y$ if f is an element of $\mathrm{Hom}_{\mathscr{C}}(x, y)$.
- If $f : x \to y$, x is called the *domain* or *source* of f and y the *codomain* or *target* of f.
- For our purpose, it is sufficient to consider that for every $(x, y) \in \mathrm{Ob}(\mathscr{C})^2$, $\mathrm{Hom}_{\mathscr{C}}(x, y)$ is a set. In this case, the category is said to be *locally small.*

Example II.2.

We only give examples that will be useful in this text. For an extensive list of examples, see [Chapter 1, Examples 2.2].

1) $\mathscr{S}et$, $\mathscr{A}b$, $\mathscr{R}ings$ will respectively denote the category of sets, abelian groups, commutative

rings with unit, with clearly associated morphisms.

2) Let $(X, \mathcal{T})$ be a topological space. We define its topology category $\mathcal{T}_X$ having objects $\mathrm{Ob}(\mathcal{T}_X) = \mathcal{T}$ and morphism $\mathrm{Hom}_{\mathcal{T}_X}(U,V) = \begin{cases} U \hookrightarrow V & \text{if } U \subseteq V \\ \emptyset & \text{otherwise.} \end{cases}$

Definition II.2.

Let $\mathscr{C}$ be a category.
A category $\mathscr{D}$ is said to be a *subcategory* of $\mathscr{C}$ if the following are satisfied :

1. every object of $\mathscr{D}$ is an object of $\mathscr{C}$,
2. for every $(x,y) \in \mathrm{Ob}(\mathscr{D})^2$, then $\mathrm{Hom}_{\mathscr{D}}(x,y) \subseteq \mathrm{Hom}_{\mathscr{C}}(x,y)$,
3. for every $x \in \mathrm{Ob}(\mathscr{D})$, then $1_x \in \mathrm{Hom}_{\mathscr{D}}(x,x)$,
4. for every $(x,y,z) \in \mathrm{Ob}(\mathscr{D})^3$, every $(f,g) \in \mathrm{Hom}_{\mathscr{D}}(x,y) \times \mathrm{Hom}_{\mathscr{D}}(y,z)$, then $g \circ f \in \mathrm{Hom}_{\mathscr{D}}(x,z)$.

$\mathscr{D}$ is said to be a *full subcategory* of $\mathscr{C}$ if in addition, for every $(x,y) \in \mathrm{Ob}(\mathscr{D})^2$, then $\mathrm{Hom}_{\mathscr{D}}(x,y) = \mathrm{Hom}_{\mathscr{C}}(x,y)$.

b. Functors

c. Natural transformations

II.1.2 (Co-)limits

a. (Co-)kernels

Definition II.3.

Let $\mathscr{C}$ be an additive category, $f : X \to Y$ a morphism.

1) A *kernel* of f is a couple (U,u) with $U \in \mathrm{Ob}(\mathscr{C})$ and $u : U \to X$ such that :
   a. $fu = 0$,
   b. for every couple (U',u') with $U' \in \mathrm{Ob}(\mathscr{C})$ and $u' : U' \to X$ satisfying $fu' = 0$, there exists a unique $g : U' \to U$ so that $ug = u'$:

$$\begin{tikzcd} U \arrow[r, "u"] & X \arrow[r, "f"] & Y. \\ U' \arrow[u, dashed, "\exists! g"] \arrow[ur, "u'"'] & & \end{tikzcd}$$

2) A *cokernel* of f is a couple (V,v) with $V \in \mathrm{Ob}(\mathscr{C})$ and $v : Y \to V$ such that :
   a. $vf = 0$,
   b. for every couple (V',v') with $V' \in \mathrm{Ob}(\mathscr{C})$ and $v' : V' \to X$ satisfying $v'f = 0$, there exists a unique $g : V \to V'$ so that $gv = v'$:

$$\begin{tikzcd} X \arrow[r, "f"] & Y \arrow[r, "v"] \arrow[dr, "v'"'] & V \arrow[d, dashed, "\exists! g"] \\ & & V'. \end{tikzcd}$$

b. Canonical factorisation

See [Ass22, VII §4].

Definition II.4.

Let $\mathscr{C}$ be an additive category and f a morphism in $\mathscr{C}$. We define :

1) the image of f as the kernel of the cokernel of f,
2) the coimage of f as the cokernel of the kernel of f.

Proposition II.1 ([Ass22], Proposition 4.3).

Let $\mathscr{C}$ be an additive category in which every morphism admits a kernel and a cokernel. Let $f : x \to y$ be a morphism.
Then, there exists a unique morphism $\bar{f} : \mathrm{Coim}(f) \to \mathrm{Im}(f)$ so that $f = j\bar{f}p$, meaning that the following diagram commutes :

$$\begin{tikzcd}
\mathrm{Ker}(f) \arrow[r, "k"] & X \arrow[r, "f"] \arrow[d, two heads, "p"'] & Y \arrow[r, "c"] & \mathrm{Coker}(f) \\
& \mathrm{Coim}(f) \arrow[r, dashed, "\exists!\,\bar{f}"] & \mathrm{Im}(f) \arrow[u, hook, "j"'] &
\end{tikzcd}$$

This decomposition is called the *canonical factorisation* of f.

II.1.3 Some special types of categories

a. Additive categories

See [Ass22, VI §2, VII §2].

Definition II.5.

A category $\mathscr{C}$ is said to be a *additive* if :

1. for every $x, y \in \mathrm{Ob}(\mathscr{C})$, the set $\mathrm{Hom}_{\mathscr{C}}(x, y)$ is a abelian group,
2. the composition is left and right distributive over the sum,
3. it admits finite products and coproducts.

b. Abelian categories

See [Ass22, VIII §2, 3].

Definition II.6.

Let $\mathscr{C}$ be a category, so that every morphism admits kernel and cokernel, and f a morphism. f is said to be *strict* if its canonical factorisation $\bar{f}$ (see II.1) is an isomorphism.

Proposition II.2 ([Ass22], Theorem 2.8).

With the same notation as the previous definition, the morphism f is strict if, and only if there exist a kernel j and a cokernel p such that $f = jp$.

One can show that in this case, $j = \mathrm{Im}(f)$ and $p = \mathrm{Coim}(f)$ (see Corollary 2.10 in [Ass22]).

Corollary II.3 ([Ass22], Corollary 2.9).

A strict morphism f that is an epimorphism and a monomorphism is an isomorphism.

Definition II.7.

An additive category is said to be *abelian* if :

1. every morphism admits a kernel and a cokernel,
2. every morphism is strict.

II.1.4 Homology, cohomology, homotopy

The categorical approach to (co)homology have historically been motivated by algebraic topology and geometry. It is entirely normal for the notation to remind the reader of concepts already encountered in applications.
We follow [Ass22, IX§5].
In this subsubsection, $\mathscr{C}$ is an abelian category.

a. First definitions

Definition II.8.

1) A complex of chains is the data of $C_\bullet = (C_n, d_n)_{n\in\mathbb{Z}}$ with the C_n's objects of $\mathscr{C}$ and $d_n : C_n \to C_{n-1}$ such that for every $n \in \mathbb{Z}, d_n \circ d_{n+1} = 0$. We will denote :

$$C_\bullet : \dots \to C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \to \dots$$

2) A complex of cochains is the data of $C^\bullet = (C^n, d^n)_{n\in\mathbb{Z}}$ with the C^n's objects of $\mathscr{C}$ and $d^n : C^n \to C^{n+1}$ such that for every $n \in \mathbb{Z}, d^n \circ d^{n-1} = 0$. We will denote :

$$C^\bullet : \dots \to C^{n-1} \xrightarrow{d^{n-1}} C^n \xrightarrow{d^n} C^{n+1} \to \dots$$

The rest of the text will mainly use the second definition, so we will state everything in this term, but the dual can easily be found by the reader.

Remark II.3.

The condition $d^n \circ d^{n-1} = 0$ means $\mathrm{Im}(d^{n-1}) \subseteq Ker(d^n)$ (when this notation makes sense in $\mathscr{C}$).

Definition II.9.

Let $C^\bullet$ be a complex of cochains.
$C^\bullet$ is said to be exact if for every $n \in \mathbb{Z}$ the following holds : $\mathrm{Ker}(d^n) = \mathrm{Im}(d^{n-1})$.

Example II.4.

Let $C^\bullet$ be a complex of cochains and $n \in \mathbb{Z}$. We say that $C^\bullet$ is *concentrated in degree n* if

$C_m = 0$ for $n \neq m$. It is exact if, and only if, C_m is also 0.

Remark II.5.

By abuse of notations, if the context if clear, we will denote all the applications between cochains d^n.

Definition II.10.

Let $C^\bullet$ and $D^\bullet$ two complexes of cochains.
A morphism $f^\bullet : C^\bullet \to C'^\bullet$ is a sequence of morphism $(f^n : C^n \to C'^n)_{n\in\mathbb{Z}}$ such that for every $n \in \mathbb{Z}$, $f^n \circ d^{n-1} = d'^n \circ f^{n-1}$, *ie* the following diagram commutes :

$$\begin{tikzcd} \dots \arrow[r] & C^{n-1} \arrow[r, "d^{n-1}"] \arrow[d, "f^{n-1}"] & C^n \arrow[r, "d^n"] \arrow[d, "f^n"] & C^{n+1} \arrow[r] \arrow[d, "f^{n+1}"] & \dots \\ \dots \arrow[r] & C'^{n-1} \arrow[r, "d'^{n-1}"] & C'^n \arrow[r, "d'^n"] & C'^{n+1} \arrow[r] & \dots . \end{tikzcd}$$

With this construction, complexes of cochains form a category, denoted by $\mathrm{Comp}^\bullet(\mathscr{C})$.

b. Cocycle, coboundary, cohomology

Definition II.11.

Let $C^\bullet$ be a complex of cochains.

1) The objects $Z^n(C^\bullet) := \mathrm{Ker}(d^n)$ are called *n-cocycles* of the complex,
2) The objects $B^n(C^\bullet) := \mathrm{Im}(d^{n-1})$ are called *n-coboundaries* of the complex,
1) The objects $H^n(C^\bullet) := Z^n(C^\bullet)/B^n(C^\bullet)$ are called *n-cohomologies* of the complex.

Lemma II.4.

Z^n, B^n and H^n induces covariant functors on $\mathrm{Comp}^\bullet(\mathscr{C})$, and there is an exact short sequence of functors :

$$0 \to B^n \to Z^n \to H^n \to 0.$$

Definition II.12.

H^n is called the *n-th cohomology functor*.

The cohomology objects are a mean to measure the exactness default of a cochain at a certain degree : $H^n(C^\bullet) = 0$ if, and only if $C^\bullet$ is exact at degree n.
The cohomology functor is not exact (meaning, it does not transform a short exact sequence in another one) in general, but gives rise to a *long exact sequence in cohomology* :

Theorem II.5 ([Ass22], Theorem 5.12).

Let $0 \to C'^\bullet \xrightarrow{f^\bullet} C^\bullet \xrightarrow{g^\bullet} C''^\bullet \to 0$ a short exact sequence of complexes of cochains. There exists an exact sequence :

$$\cdots \to H^n(C'^\bullet) \xrightarrow{H^n(f^\bullet)} H^n(C^\bullet) \xrightarrow{H^n(g^\bullet)} H^n(C''^\bullet) \xrightarrow{\delta^n} H^{n+1}(C'^\bullet) \xrightarrow{H^{n+1}(f^\bullet)} H^{n+1}(C'^\bullet) \to \dots$$

c. Homotopy

Two complexes of cochains can have the same n-th cohomology object (for all n) without being isomorphic, and two morphisms can induce the same in the cohomology without beoing equal. An important relation between morphisms having inducing the same in cohomology is the *homotopy* :

Definition II.13.

Let $f^\bullet, g^\bullet : C^\bullet \to C'^\bullet$ two morphisms of complexes of cochains.
$f^\bullet$ and $g^\bullet$ are called homotopic if there exists a sequence $(s^n : C^n \to C'^{n-1})_{n\in\mathbb{Z}}$ such that for all $n \in \mathbb{Z}$, $f^n - g^n = s^{n+1}d^n + d'^{n-1}s^n$:

$$\begin{tikzcd}
\dots \arrow[r] & C^{n-1} \arrow[r, "d^{n-1}"] \arrow[d, shift right=2, "g^{n-1}"'] \arrow[d, shift left=2, "f^{n-1}"] & C^n \arrow[r, "d^n"] \arrow[d, shift right=2, "g^n"'] \arrow[d, shift left=2, "f^n"] \arrow[dl, dashed, "s^n"'] & C^{n+1} \arrow[r] \arrow[d, shift right=2, "g^{n+1}"'] \arrow[d, shift left=2, "f^{n+1}"] \arrow[dl, dashed, "s^{n+1}"'] & \dots \\
\dots \arrow[r] & C'^{n-1} \arrow[r, "d'^{n-1}"'] & C'^n \arrow[r, "d'^n"'] & C'^{n+1} \arrow[r] & \dots
\end{tikzcd}$$

Lemma II.6 ([Ass22], Lemma 5.14).

The homotopic relation on $\mathrm{Hom}_{\mathrm{Comp}^\bullet(\mathscr{C})}(C^\bullet, C'^\bullet)$ is an equivalence relation. Moreover, if $f^\bullet$ and $g^\bullet$ are homotopic, then $H^n(f^\bullet) = H^n(g^\bullet)$ for every $n \in \mathbb{Z}$.

II.1.5 Derived categories

II.2 Sheaf theory

In the following, X is a topological space and $\mathcal{T}_X$ will denote its topology category.

Definition II.14.

Let $\mathscr{C}$ be a category (often $\mathscr{C} = \mathscr{S}et, \mathscr{A}b, \mathscr{R}ings$...).
A *presheaf* of $\mathscr{C}$ is a contravariant functor $\mathcal{F} : \mathcal{T}_X \to \mathscr{C}$.
Explicitly, it is the data of :

1. for every open set U of X, an object $\mathcal{F}(U)$ of $\mathscr{C}$,
2. for every couple of open sets $V \subseteq U$ of X, a morphism $\mathrm{res}_{U,V} : \mathcal{F}(V) \to \mathcal{F}(U)$ verifying :
    a. for every open set U of X, then $\mathrm{res}_{U,U} = \mathrm{id}_U$,
    b. for every open sets $U \subseteq V \subseteq W$ of X, then $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

The objects $\mathcal{F}(U)$ are often called *sections of $\mathcal{F}$ over U*, and for avoiding confusion, will we denoted $\Gamma(U, \mathcal{F})$.

Remark II.6.

If the context is clear, we will denote $\mathrm{res}_{U,V}(f) = f|_U$ (the notation is justified in the next lines).

Example II.7.

1) The sheaf of real valued continuous functions $C^0(-,\mathbb{R})$ with classical restriction functions $\mathrm{res}(U,V)(f) = f|_U$.
2) If X has an additional structure, the above example can be further specified : if X is a differentiable manifold, we can consider $C^\infty(-,\mathbb{R})$, if X is a complex manifold, $\mathscr{O}(-,\mathbb{C})$ ...
3) If C is an object of $\mathscr{C}$, the constant presheaf $\underline{C}$. We will be intersted in $\underline{\mathbb{Z}}$ or $\underline{\mathbb{C}}$.
4) The sheaf of real valued bounded functions $B(-,\mathbb{R})$.

Presheaves comes from a category point of view, so we need morphisms :

Definition II.15.

A *morphism of presheaves* (over the same space X) $\varphi : \mathcal{F} \to \mathcal{G}$ is a natural transformation. More explicitly, it is the data, for each $U \in \mathcal{T}_X$, of a morphism $\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$ of $\mathscr{C}$ such that the following diagram commutes for every $U \subseteq V$:

$$\begin{tikzcd} \mathcal{F}(V) \arrow[r, "\varphi_V"] \arrow[d, "\mathrm{res}^{\mathcal{F}}_{U,V}"'] & \mathcal{G}(V) \arrow[d, "\mathrm{res}^{\mathcal{G}}_{U,V}"] \\ \mathcal{F}(U) \arrow[r, "\varphi_U"] & \mathcal{G}(U) \end{tikzcd}$$

Definition II.16.

A *sheaf* $\mathcal{F}$ is a presheaf satisfying the extra conditions for any open cover $\{U_i, i \in I\}$ of an open set U of X:

3. for every $f, g \in \mathcal{F}(U)$ such that : $\forall i \in I,\ f|_{U_i} = g|_{U_i}$, then $f = g$.
4. for every $(f_i) \in \prod_{i\in I} \mathcal{F}(U_i)$ such that : $\forall (i,j) \in I^2,\ f_i|_{U_i\cap U_j} = f_j|_{U_i\cap U_j}$, then there exists $f \in \mathcal{F}(U)$ with $f|_{U_i} = f_i$ for any $i \in I$.

Remark II.8.

For $U = \emptyset$, it says that $\mathcal{F}(\emptyset)$ is the terminal object of $\mathscr{C}$ (if such one exists). Indeed, we can take the open covering of $\emptyset$ by a family indexed by $\emptyset$ itself, and we then have an arrow $\mathcal{F}(\emptyset) \to$

Example II.9.

- 1) and 2) are indeed sheaves.
- 3) is not, because of the previous remark.
- 4) is not either, with the conter example $\mathbb{R} = \bigcup_{n\in\mathbb{N}} (-n,n)$ and $f : x \mapsto x$.
- If X is Hausdorff, x_* is a point of X, (S,s) a pointed set, the *skycraper sheaf* is given by $\mathrm{sky}_{x_*}(S) : U \mapsto \begin{cases} S & \text{if } x_* \in U \\ s & \text{otherwise.} \end{cases}$

Definition II.17.

We will denote by PSh_X, respectively Sh_X, the category of presheaves, respectively the full subcategory of sheaves, over X.

One can ask what looks like the behavior of the sheaf very locally, on a point. Since a singleton is in general not a point, we need the following definition.

Definition II.18.

Let $\mathcal{F}$ be a (pre)sheaf.
We define the *stalk* at $x \in X$ as :

$$\mathcal{F}_x := \{(U,s) |\ x \in U,\ U \in \mathcal{T}_X,\ s \in \mathcal{F}(U)\} \big/ \sim ,$$

where $(U,s) \sim (V,t) \Leftrightarrow (\exists W \subset U \cap V) : s|_W = t|_W$.

Remark II.10.

From a categorical point of view, it is also the direct limit : $\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U)$, where the limit is taken over open set U.

It turns out that this localization principle can be used to transform a presheaf into a sheaf in a natural way (in the sense of what follows).

Definition II.19.

Let $\mathcal{F}$ be a presheaf.
We define for every open set U

$$\mathcal{F}^\sharp(U) = \left\{ s : U \to \coprod_{x \in U} \mathcal{F}_x \ \middle|\ \left\{ \begin{array}{l} \forall x \in U,\ s(x) \in \mathcal{F}_x \\ \exists V \subset U, \exists t \in \mathcal{F}(V) \text{ s.th.} \forall y \in V,\ s(y) = t(y) \end{array} \right. \right\}.$$

$\mathcal{F}^\sharp$ is called the *sheafification of* $\mathcal{F}$.

Proposition II.7.

$\mathcal{F}^\sharp$ is a sheaf.

Example II.11.

$\underline{\mathbb{R}}^\sharp$ is the sheaf of locally constant functions, more precisely the sheaf of continuous functions $C^0(-, \mathbb{R})$, where $\mathbb{R}$ is equipped with the discrete topology.

Proposition II.8.

$((-)^\sharp, \mathrm{Forget})$ is an adjoint pair of functors between PSh_X and Sh_X.

Proof

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□

Example II.12.

Suppose that $\mathscr{C}$ admits kernels and cokernels (this assumption holds whenever kernels and cokernels are written).
For $\varphi : \mathcal{F} \to \mathcal{G}$, we define $\mathrm{Ker}(\varphi)(U) := \mathrm{Ker}(\varphi_U)$, $\mathrm{Im}(\varphi)(U) := \mathrm{Im}(\varphi_U)$ and $\mathrm{Coker}(\varphi)(U) := \mathrm{Coker}(\varphi_U)$.
The first one is always a sheaf, but the two others are not in general. In the following, $\mathrm{Im}(\varphi)$ or $\mathrm{Coker}(\varphi)$ will always denote the sheafification of these presheaves.

Proposition II.9.

A complex $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$ is exact if, and only if the induced complex on stalks $0 \to \mathcal{F}_x \to \mathcal{G}_x \to \mathcal{H}_x \to 0$ is exact for every $x \in X$.

Proof

.

□

Remark II.13.

In general, it is not equivalent of the exactness of the induced sequence over every open set U. One could take as a counterexample the exponential sequence on $\mathbb{C}$:

$$\begin{array}{ccccccccc} 0 & \to & \underline{\mathbb{Z}} & \to & \mathscr{O}_{\mathbb{C}} & \to & \mathscr{O}^*_{\mathbb{C}} & \to & 0, \\ & & & & f & \mapsto & \exp(2\pi i f) & & \end{array}$$

that is not exact over $\mathbb{C}^*$, because the map $\mathscr{O}_{\mathbb{C}^*} \to \mathscr{O}^*_{\mathbb{C}^*}$ is not surjective : the map $z \mapsto z$ does not belong to the image since log is not well defined over $\mathbb{C}^*$.
This default in the exactness of these short sequences actually gives rise to the important theory of *sheaf cohomology*, see II.5.1.

Proposition II.10.

Let $X = \cup_{i \in I} U_i$ an open covering of X. For every $i \in I$, let $\mathcal{F}_i$ a sheaf on U_i.
Assume that for every $i \neq j \in I$, there exists an isomorphism

$$\theta_{i,j} : \mathcal{F}_i|_{U_i \cap U_j} \xrightarrow{\sim} \mathcal{F}_j|_{U_i \cap U_j},$$

satisfying the condition $\theta_{j,k} \circ \theta_{i,j} = \theta_{i,k}$ on $U_i \cap U_j \cap U_k$ for every triplet of distinct elements of I.
Then, there exists a sheaf $\mathcal{F}$ on X such that for every $i \in I, \mathcal{F}|_{U_i} = \mathcal{F}_i$. Moreover, this sheaf is unique, up to unique isomorphism.

PROOF

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□

II.3 Complex geometry

II.3.1 First definitions

Definition II.20.

Let X be a differentiable manifold of dimension $2n$.
An *holomorphic atlas* is a set $\{(U_i, \varphi_i), i \in I\}$ where $X = \bigcup_{i \in I} U_i$ is an open cover and $\varphi : U_i \to \mathbb{C}^n$ are homeomorphism on their image, such that for all $(i,j) \in I^2$, the *transition map* $\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$ is holomorphic.

Such a couple (U_i, φ_i) is called a *local coordinate system*, or just *coordinates*.

We say that two holomorpic atlases $\{(U_i, \varphi_i), i \in I\}$ and $\{(\tilde{U}_j, \tilde{\varphi}_j), j \in J\}$ are *equivalent* if for all $(i,j) \in I \times J, \tilde{\varphi}_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap \tilde{U}_j) \to \tilde{\varphi}_j(U_i \cap \tilde{U}_j)$.

Definition II.21.

A *complex manifold of dimension* n is a real manifold X of dimension $2n$ endowed with an equivalence class of holomorphic atlases.

Remark II.14.

In the following, unless explicitly stated otherwise, the complex manifold will be endowed with an representant of the equivalente class, denoted by $\{(U_i, \varphi_i), i \in I\}$.

Definition II.22.

Let X be a complex manifold and $f : X \to \mathbb{C}$.
We say that f is *holomorphic* if for all local coordinates (U, φ), $f \circ \varphi : \varphi(U) \to \mathbb{C}$ is holomorphic.

Definition II.23.

Let X be a complex manifold.
We denote by $\mathscr{O}_X$ the sheaf of holomorphic function on X, *ie* $\mathscr{O}_X : U \mapsto \{f : U \to \mathbb{C}, f \text{ holomorphic}\}$.

Remark II.15.

$\mathscr{O}_{X,x}$ will denote the stalk of $\mathscr{O}_X$ at x. It coincides with the germs of holomorphic functions at x. *Via* local coordinates (U, φ), with $\varphi(x_*) = 0$ for some $x_* \in U$, we have an isomorphism $\mathscr{O}_{X,x} \simeq \mathscr{O}_{\mathbb{C}^n,0}$. It is an integral domain, and we denote by $Q(\mathscr{O}_{X,x})$ the quotient field associated.

More generally, if the codomain is also a complex manifold, we have the following definition :

Definition II.24.

Let X and Y be complex manifolds and $f : X \to Y$ continuous.
We say that f is *holomorphic* if for all local coordinates (U, φ) on X and (U', φ') on Y,

$\varphi' \circ f \circ \varphi : \varphi(U \cap f^{-1}(U')) \to \varphi'(U')$ is holomorphic.

Definition II.25.

Let X be a complex manifold.
A *meromorphic function* on X is a map $f : X \to \bigsqcup_{x \in X} Q(\mathscr{O}_{X,x})$ so that, for all $x_* \in X$, there exists an open neighborhood $U \subset X$ of x_* and $g, h \in \mathscr{O}_U$ verifying : $(\forall x \in U),\ f_x = \frac{g}{h}$.
The sheaf of meromorphic functions is denoted $\mathcal{K}_X$.

Remark II.16.

If $U \subset X$ is an connected open subset, then $\mathcal{K}_X(U)$ is a field, and if X itself is connected, $\mathcal{K}(X) = \mathcal{K}_X(X)$ is called the function field of X.

II.3.2 Vector bundle

Definition II.26.

Let X be a complex manifold and r a positive integer.
A *holomorphic vector bundle* or rank r on X is a complex manifold E equipped with a holomorphic map $\pi : E \to X$ such that :

1. for all $p \in X$, the *fiber* $E_p := \pi^{-1}(p)$ is a complex vector space of dimension r,
2. there exists an open covering $X = \bigcup_{i \in I} U_i$ and biholomorphic maps $\psi_i : \pi^{-1}(U_i) \to U_i \times \mathbb{C}^r$, called the *trivialization maps*, satisfying :
   a. the following diagram commutes :

$$\begin{tikzcd} \pi^{-1}(U_i) \arrow[rr, "\psi_i"] \arrow[dr, "\pi"'] & & U_i \times \mathbb{C}^r \arrow[dl, "\mathrm{pr}_1"] \\ & U_i & \end{tikzcd}$$

   b. The induced map $E_p \to \mathbb{C}^r$ for any $p \in U_i$ is $\mathbb{C}$-linear.

If $r = 1$, we say that (E, π) is a *line bundle*.

Remark II.17.

The travialization maps induce linear isomorphisms $\psi_{i,j} = \psi_i \circ \psi_j^{-1} : U_i \cap U_j \to \mathrm{GL}_r(\mathbb{C})$, and the collection $\{\psi_{i,j}, i, j \in I\}$ is called a cocycle.
In particular, it verifies the conditions :

i. $\psi_{i,i} = \mathrm{id}_{U_i}$,
ii. $\psi_{j,k} \circ \psi_{i,j} = \psi_{i,k}$.

Proposition II.11.

Let X be a complex manifold and let r be a positive integer.
The categories $\mathrm{VectBun}^r(X)$ and $\mathrm{LocFree}_r(X)$ are equivalent.

Proof

We define

$$\mathcal{F}: \begin{array}{ccc} \mathrm{VectBun}_r(X) & \longrightarrow & \mathrm{LocFree}_r(X) \\ E & \longmapsto & \mathcal{F}_E \end{array},$$

where $\mathcal{F}_E$ is the sheaf of sections of E. It is well defined by ??.
It is a functor : given a morphism of vector bundles $f : E_1 \to E_2$, we define for each open set U of X the function

$$\varphi_U: \begin{array}{ccc} \mathcal{F}_1(U) & \longrightarrow & \mathcal{F}_2(U) \\ s & \longmapsto & f \circ s \end{array}.$$

φ clearly defines a morphism of sheaves.
In order to construct a quasi-inverse, let $\mathcal{F} \in \mathcal{S}_X$ and $\psi : \mathcal{F}(U_i) \to \mathscr{O}_{U_i}{}^{\oplus r}$ local trivialisations on an open cover $\{U_i, i \in I\}$ of X.

□

II.3.3 Connections

Definition II.27.

Let $\mathcal{E}$ be a locally free $\mathscr{O}_X$-modules.
A *connection on* $\mathcal{E}$ is a $\mathbb{C}$-linear application $\nabla : \mathcal{E} \to \mathcal{E} \otimes_{\mathscr{O}_X} \Omega^1_X$ satisfying the *Liebniz relation* : for any local sections s and local function f, then

$$\nabla(fs) = df \otimes s + f\nabla(s).$$

In particular, it induces, for each integer $k > 0$, a $\mathbb{C}$-linear map

$$\nabla^{(k)}: \begin{array}{ccc} \mathcal{E} \otimes_{\mathscr{O}_X} \Omega^k_X & \longrightarrow & \mathcal{E} \otimes_{\mathscr{O}_X} \Omega^{k+1}_X \\ s \otimes \omega & \longmapsto & (-1)^k \nabla s \wedge \omega + s \otimes dw \end{array}$$

A connection ∇ is said to be *flat* (or *integrable*) if the *curvature* $K := \nabla^{(1)} \circ \nabla$ is zero.

Remark II.18.

In dimension 1, we have $\Omega^2_X = 0$, so all connections are flat.

Example II.19.

1) d is a connection on $\mathscr{O}_X$ by identifying $\mathscr{O}_X \otimes_{\mathscr{O}_X} \Omega^1_X \simeq \Omega^1_X$. It induces a connection on $\mathscr{O}^r_X$ acting coordinate-wise.
2) For every $\omega \in \Omega^1_X$, the map

$$d + \omega: \begin{array}{ccc} \mathscr{O}_X & \longrightarrow & \Omega^1_X \\ f & \longmapsto & df + \omega f \end{array}$$

is a connection. More generally, if ω is a $r \times r$ matrix with coefficients in Ω^1_X, the map $d + \omega$ is a connection on $\mathscr{O}^r_X$.

The following result says that these are the only ones on $\mathscr{O}^r_X$, and then the only ones locally on $\mathcal{E}$.

Lemma II.12.

Every connection ∇ on $\mathscr{O}_X^r$ is written $d+\omega$ for some ω as above.

Proof

For every $f \in \mathscr{O}_X$ and $h \in \mathscr{O}_X^r$, we have by the Liebniz rule :

$$(\nabla - d)(fh) = f\nabla(h) + df \otimes h - fdh - df \otimes h = f(\nabla - d)h.$$

Thus, $\nabla - d : \mathscr{O}_X^r \to \Omega_X^1$ is $\mathscr{O}_X$-linear and it's the multiplication by a matrix ω.

□

Remark II.20.

Connections are a reformulation of (systems) of differential equations, locally on a vector bundle.
Let $\nabla = d+\omega$ a connection on $\mathscr{O}_X^r$ (it is of this form by the previous lemma). We choose local coordinates $z^1, \dots, z^n$ on X. Then we can write

$$\omega(z) = \sum_{j=1}^{n} A_i(z) dz^j.$$

For a f holomorphic on the chart, we then have :

$$\nabla f = \sum_{j=1}^{n} \left(\frac{\partial f}{\partial x^j} + A_j f \right) dz^j,$$

and $\nabla f = 0$ is equivalent to :

$$\begin{cases} \dfrac{\partial f}{\partial z^1}(z) = A_1(z) f(z), \\ \qquad \vdots \\ \dfrac{\partial f}{\partial z^n}(z) = A_n(z) f(z), \end{cases}$$

which is a system of *partial* differential equations.

Definition II.28.

We define the category $\mathrm{LocFree}^{\nabla}(X)$ with

- locally free $\mathscr{O}_X$-modules equipped with flat connections as objects,
- morphisms of $\mathscr{O}_X$-modules $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ making

$$\begin{tikzcd} \mathcal{E}_1 \arrow[r, "\nabla_1"] \arrow[d, "\varphi"] & \mathcal{E}_1 \otimes_{\mathscr{O}_X} \Omega_X^1 \arrow[d, "\varphi \otimes 1"] \\ \mathcal{E}_2 \arrow[r, "\nabla_2"] & \mathcal{E}_2 \otimes_{\mathscr{O}_X} \Omega_X^1 \end{tikzcd}$$

  commutes, as morphisms between $(\mathcal{E}_1, \nabla_1)$ and $(\mathcal{E}_2, \nabla_2)$.

Here are some algebraic construction one can obtain with connections :

Proposition II.13.

Let $(\mathcal{E}_1, \nabla_1)$ and $(\mathcal{E}_2, \nabla_2)$ two connections on X. The followings are also connections :

1) $(\mathcal{E}_1^*, (- \otimes 1) \circ \nabla_1)$,
2) $(\mathcal{E}_1 \oplus \mathcal{E}_2, \nabla_1 \oplus \nabla_2)$,
3) $(\mathcal{E}_1 \otimes_{\mathscr{O}_X} \mathcal{E}_2, \nabla_1 \otimes 1 + 1 \otimes \nabla_2)$
4) $(\mathcal{H}om_{\mathscr{O}_X}(\mathcal{E}_1, \mathcal{E}_2), \phi \mapsto (\nabla_2 \circ \phi) - (\phi \otimes 1) \circ \nabla_1)$

II.4 Algebraic geometry

II.5 Some cohomology theory

For this section, we follow [[Voi02], §4.3] and [[HM17]]

II.5.1 Sheaf cohomology

II.5.2 de Rham cohomology

III The analytic Riemann-Hilbert correspondence

III.1 Preliminary : a correspondence in algebraic topology

We follow [[Sza12], Chapter 2] for this paragraph.

III.2 Holomorphic version on a complex manifold

Some ideas of this paragraph are taken from [[Voi02], §9.2], [[Sza12], §2.7] and [[Poo25], §7.7].

Theorem III.1 (Riemann-Hilbert correspondence, complex manifold case).

Let X be a complex manifold.
The functor

$$\begin{array}{rccl} F: & \mathrm{LocFree}^{\nabla}(X) & \longrightarrow & \mathrm{LocSys}(X) \\ & (\mathcal{E}, \nabla) & \longmapsto & \mathcal{E}^{\nabla} := \mathrm{Ker}(\nabla) \end{array}$$

induces an equivalence of categories.
Its inverse is given by

$$\begin{array}{rccl} G: & \mathrm{LocSys}(X) & \longrightarrow & \mathrm{LocFree}^{\nabla}(X) \\ & \mathcal{A} & \longmapsto & (\mathscr{O}_X \otimes_{\mathbb{C}} \mathcal{A}, d \otimes 1) \end{array}$$

Proof

Step 1 : G is well defined, that is : if $\mathcal{A} \in \mathrm{LocSys}(X)$, then $(\mathcal{E}, \nabla) := (\mathcal{A} \otimes_{\mathbb{C}} \mathscr{O}_X, 1 \otimes d) \in \mathrm{LocFree}^{\nabla}(X)$.
First remark that $d \otimes 1$ is well defined by extension of $\mathscr{O}_X \to \Omega^1_X$ with the tensor product. We also use the identification

$$(\mathcal{A} \otimes_{\mathbb{C}} \mathscr{O}_X) \otimes_{\mathscr{O}_X} \Omega^1_X \simeq \mathcal{A} \otimes_{\mathbb{C}} (\mathscr{O}_X \otimes_{\mathscr{O}_X} \Omega^1_X) \simeq \mathcal{A} \otimes_{\mathbb{C}} \Omega^1_X$$

by commutativity of the tensor product.
For a small open set such that $\mathcal{A}|_U \simeq \mathbb{C}^r$, we have $(\mathcal{A} \otimes \mathscr{O}_X)|_U \simeq \mathcal{A}|_U \otimes \mathscr{O}_U \simeq \mathbb{C}^r \otimes \mathscr{O}_U \simeq \mathscr{O}_U^{\oplus r}$.
Finally, we have :

$$\nabla^{(1)} \circ \nabla(a \otimes f) = \nabla^{(1)}((a \otimes 1) \otimes df) = \nabla(a \otimes 1) \wedge df + a \otimes d(df) = (a \otimes d1) \wedge df = 0.$$

We again used that the image of $a \otimes f$ in $\mathcal{E} \otimes_{\mathscr{O}_X} \Omega^1_X$ is identified with $(a \otimes 1) \otimes df$.
Step 2 : G is a functor.
If $\phi : A \to B$ is a morphism in $\mathrm{LocSys}(X)$, we have an induced $G(\phi) = \phi \otimes 1 : A \otimes_{\mathbb{C}} \mathscr{O}_X \to B \otimes_{\mathbb{C}} \mathscr{O}_X$ that makes the following commutes :

$$\begin{tikzcd} A \otimes_{\mathbb{C}} \mathscr{O}_X \arrow[r, "\phi\otimes 1"] \arrow[d, "1\otimes d"'] & A \otimes_{\mathbb{C}} \mathscr{O}_X \arrow[d, "1\otimes d"] \\ A \otimes_{\mathbb{C}} \Omega^1_X \arrow[r, "\phi\otimes 1"] & B \otimes_{\mathbb{C}} \Omega^1_X \end{tikzcd}$$

Step 3 : F is well defined, that is : if $(\mathcal{E}, \nabla) \in \mathrm{LocFree}(X)$, then $\mathcal{E}^\nabla \in \mathrm{LocSys}(X)$.
This is the most difficult part. Let us sketch out the proof.
The result is equivalent to saying that *locally*, the solutions of our system of differential equations given by ∇ is a complex vector space, with dimension equal to the rank of the system. One can then think of the Cauchy-Lipschitz (or Picard-Lindelöf) theorem. It indeed works when X is a curve because we have locally an ordinary differential equation. If $n > 1$, this theorem cannot be applied.
There is an alternative for this type of system : the Frobenius theorem. One corollary of this theorem is the following lemma, that implies immedialety the result.

Lemma III.2.

Let $x \in X$.
There exists an open set U containing x such that the evaluation $\mathrm{ev}_x : \mathrm{Ker}(\nabla|_U) \to \pi^{-1}(x)$ is an isomorphism (of sheaves of complex vector spaces).

Step 4 : F is a functor.

If $\phi : (\mathcal{E}_1, \nabla_1) \to (\mathcal{E}_2, \nabla_2)$ a morphism in $\mathrm{LocFree}^\nabla(X)$, then it induces a morphism of sheaves $\phi : \mathrm{Ker}(\nabla_1) \to \mathcal{E}_2$, and its image is a subset of $\mathrm{Ker}(\nabla_2)$ by the relation $\nabla_2 \circ \phi = (\phi \otimes 1) \circ \nabla_1$.
Step 5 : $GF \simeq 1_{\mathrm{LocFree}^\nabla(X)}$.
Let $(\mathcal{E}, \nabla) \in \mathrm{LocFree}^\nabla(X)$. Set

$$\alpha_{(\mathcal{E},\nabla)} : \begin{array}{ccc} \mathrm{Ker}(\nabla) \otimes_{\mathbb{C}} \mathscr{O}_X & \longrightarrow \mathcal{E} \\ s \otimes f & \longmapsto fs \end{array}.$$

It is clearly well defined and for $s \otimes f \in \mathrm{Ker}(\nabla) \otimes_{\mathbb{C}} \mathscr{O}_X$, we have, using the Liebniz rule :

$$\nabla(\alpha_{(\mathcal{E},\nabla)}(s \otimes f)) = \nabla(fs) = s \otimes df = \alpha(s \otimes 1) \otimes df = (\alpha \otimes 1)((1 \otimes d)(s \otimes f)),$$

so $\alpha_{(\mathcal{E},\nabla)}$ define a morphism in the category $\mathrm{LocFree}^\nabla(X)$.
Better, it defines a natural transformation. Let $\phi : (\mathcal{E}_1, \nabla_1) \to (\mathcal{E}_2, \nabla_2)$ a morphism in $\mathrm{LocFree}^\nabla(X)$. Then,

$$\phi(\alpha_{(\mathcal{E}_1,\nabla_1)}(f \otimes s)) = \phi(fs)$$

and

$$\alpha_{(\mathcal{E}_2,\nabla_2)}(GF(\phi)) = \alpha_{(\mathcal{E}_2,\nabla_2)}(F(\phi)(s) \otimes f) = f\phi(s).$$

This is equal because ϕ is a morphism of $\mathscr{O}_X$-modules.
To show that this is an isomorphism, let U be a small open neighborhood of x, such that $\mathrm{Ker}(\nabla|_U) \simeq \mathbb{C}^r$. Let $(s_1, \dots, s_r)$ a basis of $\mathrm{Ker}(\nabla|_U)$. Then, $(s_1 \otimes 1, \dots, s_r \otimes 1)$ is a basis of $\mathrm{Ker}(\nabla|_U) \otimes_{\mathbb{C}} \mathscr{O}_U$.
Moreover by taking U small enough, $(s_1, \dots, s_r)$ is also a $\mathscr{O}_U$-basis of $\mathcal{E}(U) \simeq \mathscr{O}_U^{\oplus r}$. Indeed, the determinant of the matrix (in any basis of $\mathcal{E}_U$) with j-th column $(s_j^{(i)})^T$ is not zero at x, so does not vanish in a small neighborhood of x.
Therefore, the matrix of $\alpha_{(\mathcal{E},\nabla)}$ in the basis $(s_1, \dots, s_r)$ (on both sides) is the identity, and is invertible.
Thus, $\alpha_{(\mathcal{E},\nabla)}$ is invertible on each stalk, and it is an isomorphism of sheaves.

Step 6 : $FG \simeq 1_{\mathrm{LocSys}(X)}$.
We clearly have for a local system $\mathcal{A}$ on X, $(\mathcal{A} \otimes_{\mathbb{C}} \mathscr{O}_X)^{1\otimes d} \simeq \mathcal{A} \otimes_{\mathbb{C}} \mathbb{C} \simeq \mathcal{A}$, and it behaves well with morphisms, such that it defines an isomorphism of functors.

□

III.3 Meromorphic version on a complex manifold

For a Riemann surface (meaning, $\dim(X) = 1$), the previous result gives us that any finite dimensional reprensentation of $\pi_1(X, x_*)$ can be obtained as the monodromy reprensentation of some linear system of holomorphic differential equations.

21st Hilbert problem : given of finite set of points

III.3.1 In dimension 1

a. Elements of Fuchs theory

We focus out study around each point of S. Without loss of generality, the case around $z = 0$ is sufficient. We will denote D_ε the disk centered at 0 of radius ε and D_ε^* the associated punctured disk at $z = 0$.

If we want to translate the definition of a connection on D_ε, we would say that such a connection is a locally free sheaf of modules $\mathcal{E}$ over the sheaf of rings of meromorphic function defined on D_ε with only a pole at 0, equipped with a map $\nabla : \mathcal{E} \to \mathcal{E} \otimes_{\mathscr{O}_{D_\varepsilon}} \Omega^1_{D_\varepsilon}$ with linearity and Liebniz propreties.
We can assume that $\mathcal{E}$ is free. Moreover, it's more convenient to work with the field of germs of meromorphic functions around 0 denoted K.
Having localised our study, we then won't work with sheaf in this paragraph. We basically follow Haefliger's lecture in [Bor+87], and the theory exposed is usually refered to as the *Fuchs theory*. This shift of persective leads us to the following definition.

Definition III.1.

A (germ of) *meromorphic connexion* at 0 is a finite dimensional vector space M over K equipped with an operator $\mathbb{C}$-linear $\nabla : M \to M$ saisfying the Liebniz rule :

$$\forall (h, u) \in K \times M : \nabla(hu) = \frac{dh}{dz} u + h\nabla u.$$

With this definition, a morphism of connection is of linear morphism commuting with connections.

Remark III.1.

The ring $D[z^{-1}]$ of germs of differential operators with meromorphic coefficients acts on a meromorphic connection (M, ∇) by $Pu = \sum_{j=1}^N a_j \nabla^j u$ if $P = \sum_{j=1}^N a_j \frac{d^j}{dz^j}$ and $u \in M$.

Remark III.2.

As explained in II.20, given a basis of M, a connection map on M is equivalent to a system of differential equations : a meromorphic connection is given by $\nabla = d - A dz$, where A is a matrix with coefficient in K, and vice versa.
A morphism between systems of equation given by matrices A and B respectively is then

a matrix $M : D_\varepsilon^* \to M_n(K)$ such that $\frac{dM}{dz} = BM - MA$, and we say that two systems of differential equations are *equivalent* if $M(z)$ is invertible for any $z \in D_\varepsilon$ (or equivalently, the associate connections are isomorphic).
In the rest of the paragraph, we will juggle between these two definitions.

Let us compute the monodromy associated to a differential system $y' = Ay$ on D_ε.
First, recall that the universal cover of D_ε^* given by the exponential is $\{t \in \mathbb{C}, |e^{2i\pi t}| < \varepsilon\}$. Multi-valued functions are thus defined on this universal cover. t and $\tilde{}$ will belong to the universal cover and z to D_ε.
Let $\tilde{S}(t) = (\tilde{U}_1, \dots, \tilde{U}_n)$ a fundamental system of multi-valued solutions. A is 1-periodic on the universal cover, so $t \mapsto \tilde{S}(t+1)$ is also a solution. This, there exists a constant invertible matrix C such that $\tilde{S}(t+1) = \tilde{S}(t)C$.

Definition III.2.

The matrix C defined above is called the *monodromy matrix* of the system.

To get back in the punctured disk, let Γ a matrix such that $C = \exp(2i\pi\Gamma)$. Then, $\tilde{S}(t)\exp(-2i\pi t\Gamma)$ is 1-periodical on the universal cover, and one can well define $\Sigma(z) = \tilde{S}(\frac{1}{2i\pi}\log(z))e^{\log(z)\Gamma}$. Finally, we set $S(z) = \Sigma(z)e^{\log(z)\Gamma}$.

One cannot expect to have a correspondence between the category of meromorphic connections and the one of representations of $\pi_1(D_\varepsilon^*)$, as the following example shows :

Example III.3.

Define the two connections $\nabla_1 = d$ and $\nabla_2 = d - \frac{dz}{z^2}$. They are not equivalent since $z \mapsto 0$ is not invertible, but they both have the trivial monodromy (the solutions of ∇_2 are of the form $Ce^{-1/z}$).

This phenomenon can be intuited as resulting of a *bad* behavior of the solutions at $z = 0$, meaning that they have there sort of an essential singularity.
Thus, we introduce a *nice* class of functions.

Definition III.3.

Let f be a multivalued function around 0.
f has *moderated growth* if, for all sector $S^\eta_{(a,b)} := \{z \in D_\eta^* \mid \arg(z) \in (a,b)\}$ over which f is defined, there exist $k \in \mathbb{N}$ and $C > 0$ such that :

$$\forall z \in S^\eta_{(a,b)},\ |f(z)| \leq C|z^{-k}|.$$

Example III.4.

1) If f is a single-valued function, f has moderate growth if, and only if f is meromorphic on D_ε (for ε small enough).
2) z^α for $\alpha \in \mathbb{C}$, $\log(z)^m$ for $m \in \mathbb{N}$, have moderate growth.

The following result if very powerful : it gives you information on the behavior of the solutions of a differential equation just by checking a condition on the coefficients of the differential operator with a very simple computation.

Theorem III.3 (Fuchs).

Let $P = \sum_{k=0}^{n} a_k(z) \frac{d^{n-k}}{dz^{n-k}}$ a differential operator on D_ε with the a_k's meromorphic.
The followings are equivalent :

i) the solutions of $Pu = 0$ have moderate growth,

ii) for every $1 \leq i \leq n$, $\frac{a_i}{a_0}$ has a pole of order at most i at 0.

These systems are those we are interested in for an expected correspondence.

Definition III.4.

A differential operator is said to be *regular* if it satifies the conditions of the previous theorem.

Example III.5.

As expected, the example given in III.3 is not regular.

Theorem III.4.

Let $(E) : y' = Ay$ a differential system on D_ε. The followings are equivalent :

i) (E) is equivalent to a regular system,

ii) (E) is equivalent to a system $y' = \frac{B(z)}{z} y$ for some matrix B holomorphic on D_ε,

iii) (E) is equivalent to a system $y' = \frac{\Gamma}{z} y$ for some constant matrix Γ.

Proof

.

□

With this theorem, the equivalent definition in terms of connections is the following :

Definition III.5.

Let (M, ∇) be a meromorphic connection.
It is said to e an *regular connection* if there exists a basis $(e_1, \ldots, e_n)$ of M over K and holomorphic functions $(b_{i,j}(z))$ such that :

$$\forall i \in [\![1, n]\!], \forall z \in D_\varepsilon,\ z\nabla e_i = -\sum_{j=1}^{n} b_{i,j}(z) e_j.$$

The following theorem is what we were aiming for.

Theorem III.5.

Two regular systems on D_ε are equivalent if, and only if, their monodromy is conjugated.

Proof

.

□

Stated like that, it is not as precise as we would have wanted for a equivalence of categories like in III.2. The next paragraph will make it more accurate, based on the theorem above.

First, a result on the good behavior of algebraic construction (see II.13) with regularity :

Proposition III.6 ([HTT08], Proposition 5.1.11).

Let M and N be regular meromorphic connections.
Then $\mathrm{Hom}_K(M,N)$ and $M \otimes_K N$ are regular.

PROOF

.

$\square$

b. The local lifting

This paragraph is partially inspired by [[Sza12], §2.7] and [[HTT08], Lemma 5.2.19] for the lifting of morphisms.
The precise Riemann-Hilbert correspondence will be stated next paragraph, based on what we already did in the holomorphic case. In this one, we try to construct a unique (regular) meromorphic connection on D_ε given a holomorphic connection on D_ε^*.
We return to the language of sheaves. Indeed, according to a general fact on non-compact Riemann surfaces, a locally free $\mathscr{O}_{D_\varepsilon^*}$-module on D_ε is actually free (for example, see [[For81], Theorem 30.4]).
We will denote by $\mathscr{O}_{D_\varepsilon}(*0)$ the sheaf of meromorphic functions on D_ε with a pole at $z=0$.

Proposition III.7.

Let $(\mathcal{E},\nabla)$ a holomorphic connection on D_ε^*.
There exists regular meromorphic connection $(\bar{\mathcal{E}},\bar{\nabla})$ on D_ε which agrees with $(\mathcal{E},\nabla)$ on D_ε.
This connection is unique up to isomorphisms (of meromorphic connections).

PROOF

We can assume $\mathcal{E} = \mathscr{O}_{D_\varepsilon}^r$ and $\nabla = d - Adz$. Set $S(z)$ a fundamental system of solutions. Let C the monodromy matrix associated to ∇, $S(z)$ a fundamental system of solutions, and let B a matrix such that $C = \exp(2i\pi B)$. We consider the connection $\bar{\nabla} = d - \frac{B}{z}dz$. A fundamental system of solutions is given by $z^B = e^{B\log(z)}$.
Let $U(z) = S(z)z^{-B}$. Then

$$U(ze^{2i\pi}) = S(ze^{2i\pi})e^{-B\log(ze^{2i\pi})} = S(z)Ce^{-2i\pi B}z^{-B} = U(z).$$

Here, by an abuse of notation, $ze^{2i\pi}$ denotes the composition by the simple loop going around 0 counter-clockwise.
Therefore, $U(z)$ is single-valued, and then holomorphic on D_ε^*. Its inverse is too, so it defines a isomorphism between $(\mathcal{E},\nabla)$ and $(\mathscr{O}_{D_\varepsilon}(*0)^r,\bar{\nabla})$.
Uniqueness comes from III.5.

$\square$

To make a functorial statement, we need additionally an extension of morphisms, we will directly do it for the global framework.

c. Global study via patching

We will state the "meromorphic" Riemann-Hilbert on $\mathbb{P}^1_{\mathbb{C}}$ minus a finite set of points. We firstly introduce some global notation compared with what we did the last two paragraphs, and then state and proof the result.
Some ideas comes from [Sza12].

In this paragraph, we set $X = \mathbb{P}^1_{\mathbb{C}}$, $D = \{s_1, \dots, s_k\} \subset X$ and $Y = X \setminus D$.
We denote by $\mathscr{O}_X(*D)$ the sheaf of meromorphic functions on X with poles along D.

Definition III.6.

Let $\bar{\mathcal{E}}$ a locally free $\mathscr{O}_X(*D)$-modules.
A *meromorphic connection* on $\bar{\mathcal{E}}$ is a $\mathbb{C}$-linear map $\bar{\nabla} : \bar{\mathcal{E}} \to \bar{\mathcal{E}} \otimes_{\mathscr{O}_X(*D)} \Omega^1_X$ satisfying the Liebniz rule.

Definition III.7.

We define the category $\mathrm{LocFree}^{\bar{\nabla}}(X, D)$ with

- locally free $\mathscr{O}_X(*D)$-modules equipped with a meromorphic (flat) connection as objects,
- morphisms of $\mathscr{O}_X(*D)$-modules commuting with the connections as morphisms.

Theorem III.8.

Let $(\mathcal{E}, \nabla)$ a holomorphic connection on Y.
There exists a regular meromorphic connection $(\bar{\mathcal{E}}, \bar{\nabla})$ on X which agrees with $(\mathcal{E}, \nabla)$ on D_ε.
This connection is unique up to isomorphisms (of meromorphic connections).

Proof

Let D_i small disks around each s_i that do not meet. On each D_i, let $(\bar{\mathcal{E}}_i, \bar{\nabla}_i)$ the unique (up to isomorphisms) lifting of $(\mathcal{E}_i, \nabla_i) := (\mathcal{E}, \nabla)|_{D_i}$ given by III.7. On an open set U containing the complementary of $\bigcup_{i=1}^k D_i$ that does not meet S, we set $\bar{\mathcal{E}} = \mathcal{E}$.
Then, we can glue back the sheaves on $X = \bigcup_{i=1}^k D_i \cup U$ by II.10 (the isomorphisms $\theta_{i,j}$ are just given by the ones coming from the construction of the $\bar{\mathcal{E}}_i$, and the triple intersection condition is empty), which gives us the sheaf $\bar{\mathcal{E}}$ on X that agrees with $\mathcal{E}$ on Y.

□

Now, we deal with morphisms :

Proposition III.9 ([HTT08], Lemma 5.2.19).

Let $\phi : (\mathcal{E}_1, \nabla_1) \to (\mathcal{E}_2, \nabla_2)$ a morphism between holomorphic connections on Y.
Then, there exists a morphism of meromorphic connections $\bar{\phi} : (\bar{\mathcal{E}}_1, \bar{\nabla}_1) \to (\bar{\mathcal{E}}_2, \bar{\nabla}_2)$ on X that restricts to ϕ.

We need two intermediate result.

Lemma III.10.

The restriction map given by III.8 induces a bijection :

$$\Gamma(X, \mathrm{Ker}(\bar{\nabla})) \xrightarrow{\sim} \Gamma(Y, \mathrm{Ker}(\nabla)).$$

PROOF

The injectivity is clear. For the surjectivity, let $s \in \Gamma(Y, \mathrm{Ker}(\nabla))$. $\bar{\nabla}$ is regular, so s, a solution of the differential equation induced is single-valued as being defined on the whole of Y, and therefore has moderate growth - meaning that it can be extended to a meromorphic section on D. This concludes.

□

This one follows the definition.

Lemma III.11.

Let $(\bar{\mathcal{E}}_1, \bar{\nabla}_1)$ and $(\bar{\mathcal{E}}_2, \bar{\nabla}_2)$ two connections on D_ε.
Then,

$$\Gamma(X, \mathcal{H}om_{\mathscr{O}_X(*D)}(\bar{\mathcal{E}}_1, \bar{\mathcal{E}}_2)^\nabla) \simeq \mathrm{Hom}_{\mathrm{LocFree}^{\bar{\nabla}}(X,D)}((\bar{\mathcal{E}}_1, \bar{\nabla}_1), (\bar{\mathcal{E}}_2, \bar{\nabla}_2)),$$

where the ∇ on the left-hand side is the one associated to the sheaf $\mathcal{H}om$ (see II.13).

PROOF (III.9)

Set $\mathcal{E} = \mathcal{H}om_{\mathscr{O}_Y}(\mathcal{E}_1, \mathcal{E}_2)$ and $\bar{\mathcal{E}} = \mathcal{H}om_{\mathscr{O}_X(*S)}(\bar{\mathcal{E}}_1, \bar{\mathcal{E}}_2)$ with the associate ∇ and $\bar{\nabla}$. By III.6, $\bar{\mathcal{E}}$ is regular, so we can apply the first lemma. The second one thus gives a bijection :

$$\mathrm{Hom}((\bar{\mathcal{E}}_1, \bar{\nabla}_1), (\bar{\mathcal{E}}_2, \bar{\nabla}_2)) \simeq \mathrm{Hom}((\mathcal{E}_1, \nabla_1), (\mathcal{E}_2, \nabla_2)),$$

the surjectivity of which ensures the result.

□

Theorem III.12.

The restriction functor

$$\mathrm{LocFree}^{\bar{\nabla}}(X, D) \to \mathrm{LocFree}^{\nabla}(Y)$$

induces an equivalence of categories.

PROOF

The pseudo-inverse functor is given by the construction above, and all the other verifications are clear.

□

With the work done the previous subsection, we have the correspondence as wished :

Corollary III.13 (Riemann-Hilbert correspondence, meromorphic case, dimension 1).

There is an equivalence of categories :

$$\mathrm{LocFree}^{\bar{\nabla}}(X, D) \cong \mathrm{LocSys}(Y).$$

III.3.2 Sketch of a proof in higher dimension

The classical reference of this subsubsection is [Del70], and the presentation by Malgrange in [[Bor+87], Part IV] and restructured in [[HTT08], §5.2].

For the general case, we work with a pair (X, D), where X is a connected complex manifold of dimension n and D a divisor of X. We set $Y = X \setminus D$.
The main goal is to build a unique regular meromorphic lifting of connection on X given a holomorphic one on Y. Notice that the central point in the work of Deligne is to redefine the regularity condition for higher dimension. Two approachs are possible : reduce to the 1-dimensional case by verifying the regularity on the pullback of the connection by some $D(0,1) \hookrightarrow X$, or to control the growth of the solutions on the associated PDE. See [[Del70], II§4, Théorème 4.1] for details.
The existence is somehow similar to what we have already done in the one-dimensional case, meaning that the can patch local lifting, and actually restrict to the case where $Y = (D^*)^r \times D^{n-r}$, where $D = D(0, \varepsilon)$, and we translate the problem in terms of monodromy representation to construct a suitable connection. This simplification for the form of Y however suits for D a *normal crossing* divisor : a divisor D is said to be of normal crossing if for every point $p \in D$, there exists local coordinates $(x_1, \dots, x_n)$ of a neighbourhood U of p in X such that $D \cap U = \{x_1 \dots x_m = 0\}$.
Fortunatly, one can always refer back to this case by the Hironaka's resolution of singularities (see [Hir64]) : there exists a map $f : X' \to X$, proper and surjective, such that $D' := f^{-1}D$ is a normal crossing divisor, and the restriction of $f : X' \setminus D' \to X \setminus D$ is an isomorphism. Therefore, the pullback of a connection M on X, g^*M, can be extended to a regular meromorphic connection on X', and can give back a regular connection on X by applying some functors (for the details, see [HTT08], Theorem 5.2.20).

The uniqueness requires more constraints, namely the log-poles of the connection around D, and the real-part of the eigenvalues associated to the connection being in a lenght-1-interval. But these constraints are actually useful to glue back local lifting, and have some control over the morphisms between connections.

IV The algebraic Riemann-Hilbert correspondence

IV.1 Elements of GAGA

IV.2 Meromorphic Riemann-Hilbert on a smooth algebraic variety

V A higher generalization : the $\mathcal{D}$-module theory

V.1 General theory of $\mathcal{D}$-modules

V.1.1 Introduction

a. First definitions

b. Relationship with PDEs

V.2 Some interesting functors

V.3 Holonomicity

VI Periods

for introduction : picard fuschs equation in a historical situation : from periods depending of a parameter, construct a differential equations attached

VI.1 Prerequisites of periods theory

VI.1.1 A first approach

VI.1.2 When geometers brought cohomology

VI.2 The Gauss-Manin connection

VI.2.1 Coming from topology

We essentially use [[Voi02], Chapter 9].

Let $\mathscr{X}$ and S be complex manifolds.

Definition VI.1.

Let $\phi : \mathscr{X} \to S$ be a holomorphic map.
We say that $\mathscr{X} \xrightarrow{\phi} S$ is a *family of complex manifold* if ϕ is a proper submersion.

We recall that proper means that the inverse image of a compact set is compact.
We will denote by $\mathscr{X}_s$ the fiber $\phi^{-1}(s)$ above $s \in S$.
The following result gives us the differentiable local behavior of such a family.

Theorem VI.1 (Ehresmann, [Voi02], Proposition 9.3).

Let $\mathscr{X} \xrightarrow{\phi} S$ be a family of complex manifold, and assume that S is contractible, with a based point O.
Then, there exists a diffeomorphism $T : \mathscr{X} \xrightarrow{\sim} \mathscr{X}_O \times S$ such that $\mathrm{pr}_2 \circ T = \phi$.

Remark VI.1.

This theorem only gives information on the differentiable structures. Indeed, there exists conter-examples for the analytic structure, see [[Poo25], Warning 4.9]. It can however be improved in this case, see [[Voi02], Proposition 9.5].

Let $\mathscr{X} \xrightarrow{\pi} S$ be a family of complex manifold. Consider the sheaf $H^q := R^q\pi_*\mathbb{C}_{\mathscr{X}}$.

VI.2.2 Some algebraic constructions

a. With the algebraic de Rham cohomology

b. Using filtrations

c. In the $\mathcal{D}$-module theory

VI.3 The variational theory of periods : interaction with Hodge theory

VII Conclusion and follow-ups

a. Irregular $\mathcal{D}$-modules and exponential periods

b. p-adic Hodge theory

c. p-adic Riemann-Hilbert correspondence

d. Arithmetic $\mathcal{D}$-modules

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